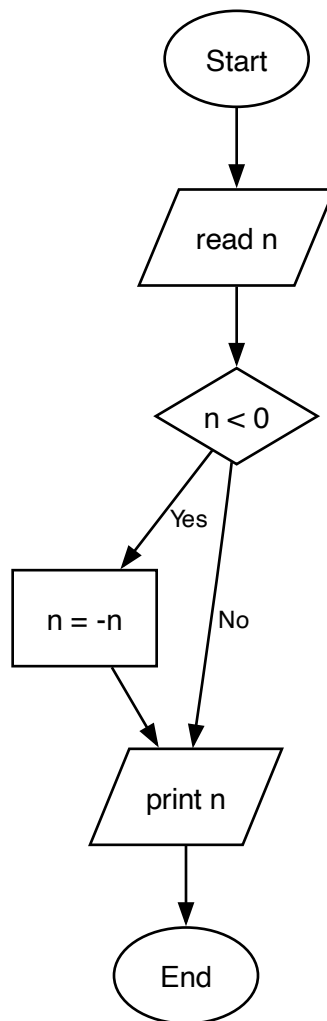


Problem 1.



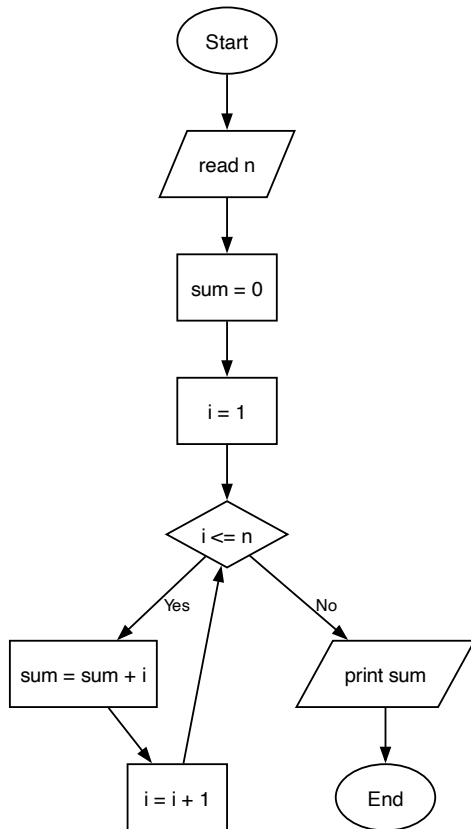
(i) For each input below, follow the flowchart and write the output.

input n	output
7	
-4	
0	
-19	

(ii) Can you tell what this algorithm is doing?

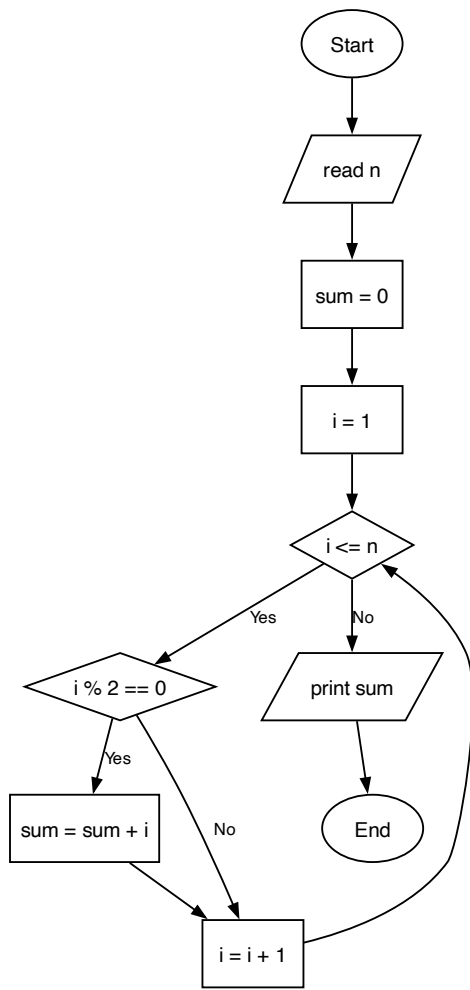
Problem 2.

Trace this algorithm for $n = 4$, and complete the trace table.

[illegible]

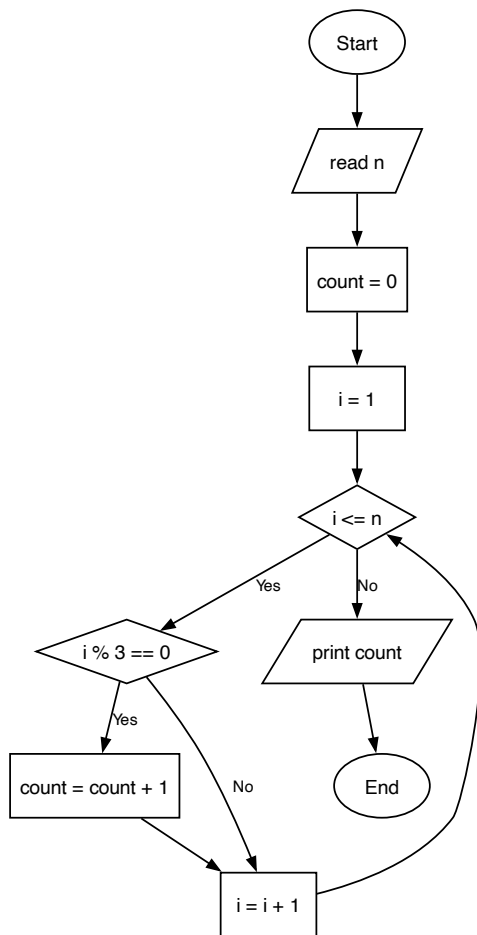
Problem 3.

Trace this algorithm for $n = 5$, and complete the trace table.

[illegible]

Problem 4.

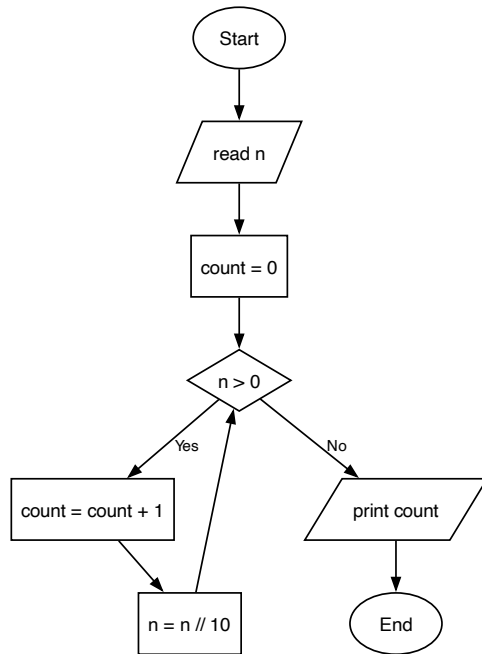
Trace this algorithm for $n = 7$, and complete the trace table.

[illegible]

Problem 5.

Trace this algorithm for $n = 472$, and complete the trace table.

Note: $n \text{ // } 10$ means whole-number division — divide and drop the remainder. For example, $472 \text{ // } 10 = 47$ (not 47.2).

[illegible]

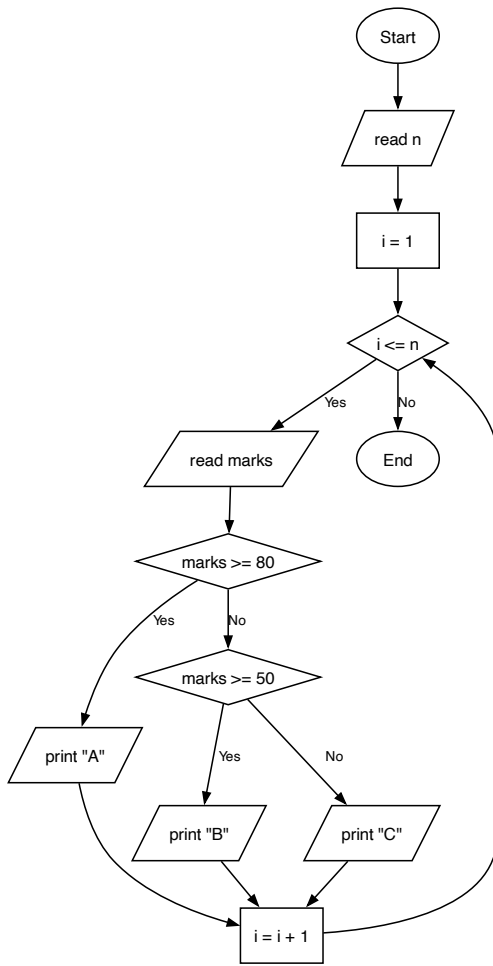
Problem 6.

Draw a flowchart for this algorithm:

Ask the user for a number n . Then go through the numbers 1, 2, 3, ... up to n , one by one: if a number is divisible by 3, print “buzz”; otherwise print the number itself.

Problem 7.

Trace this algorithm for $n = 3$, where the values of *marks* entered are 55, 82, 40 (in that order), and complete the trace table.

[illegible]